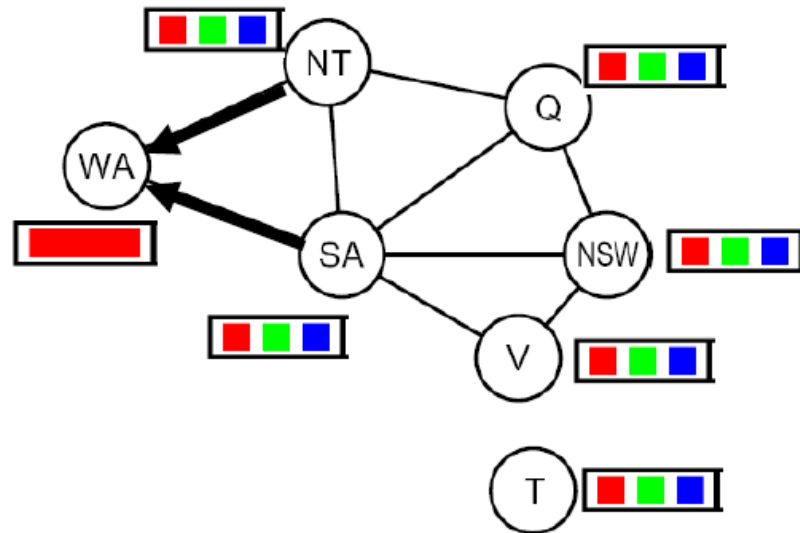
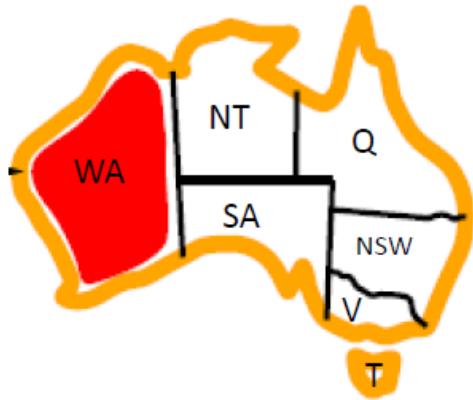


Arc – Consistency -3

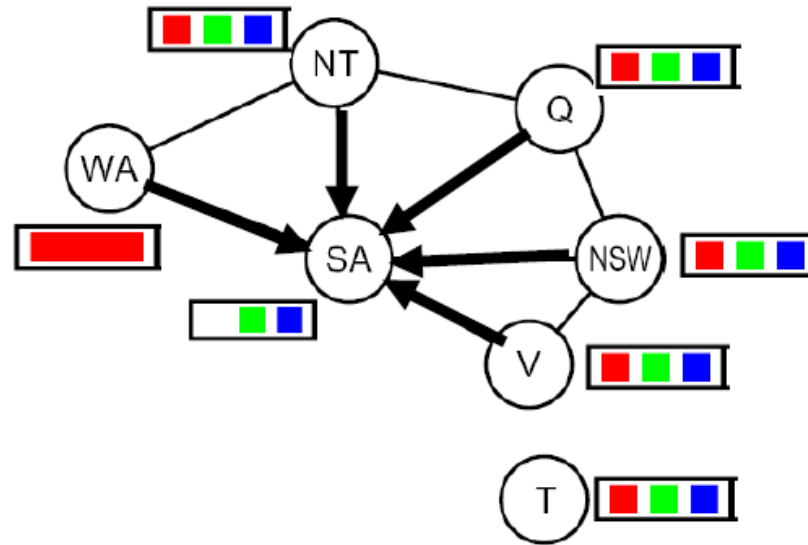
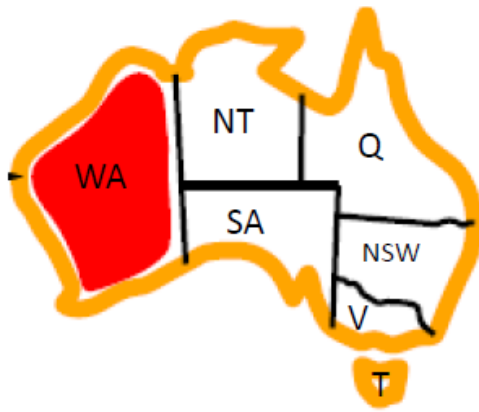
Slide credits: CMU AI, <http://ai.berkeley.edu>

AC-3: Enforce Arc Consistency of Entire CSP



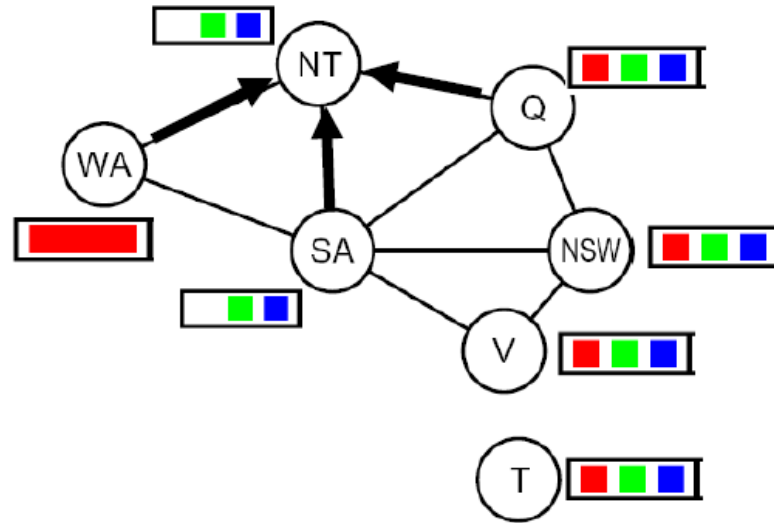
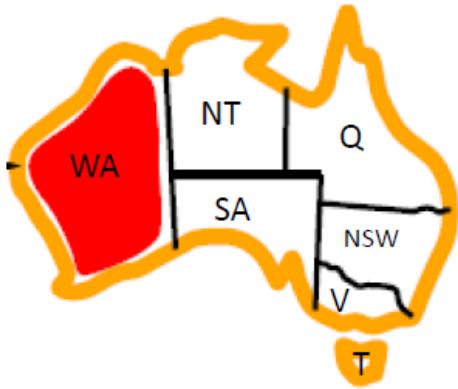
Queue:
SA->WA
NT->WA

AC-3: Enforce Arc Consistency of Entire CSP



Queue:
NT->WA
WA->SA
NT->SA
Q->SA
NSW->SA
V->SA

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

WA->SA

NT->SA

Q->SA

NSW->SA

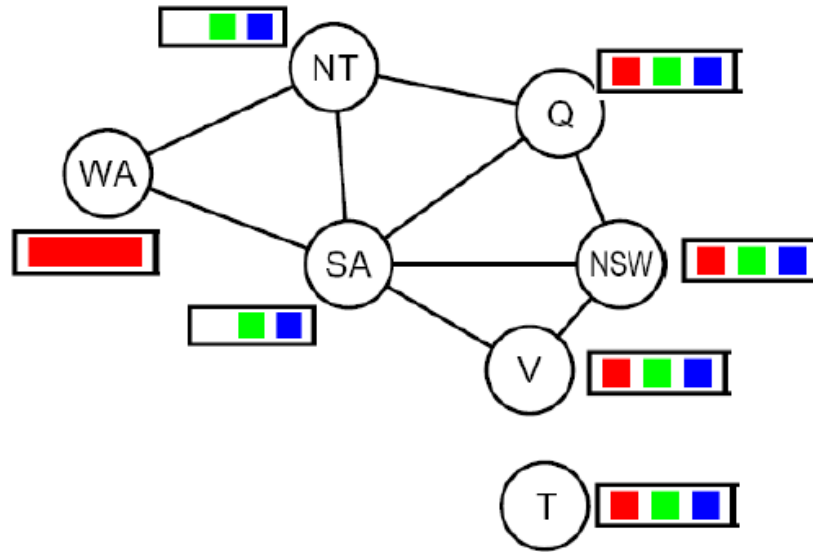
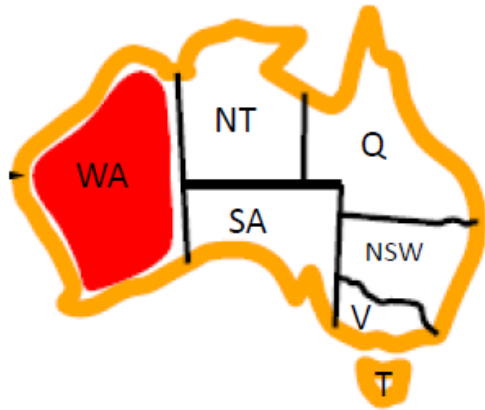
V->SA

WA->NT

SA->NT

Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

WA->SA

NT->SA

Q->SA

NSW->SA

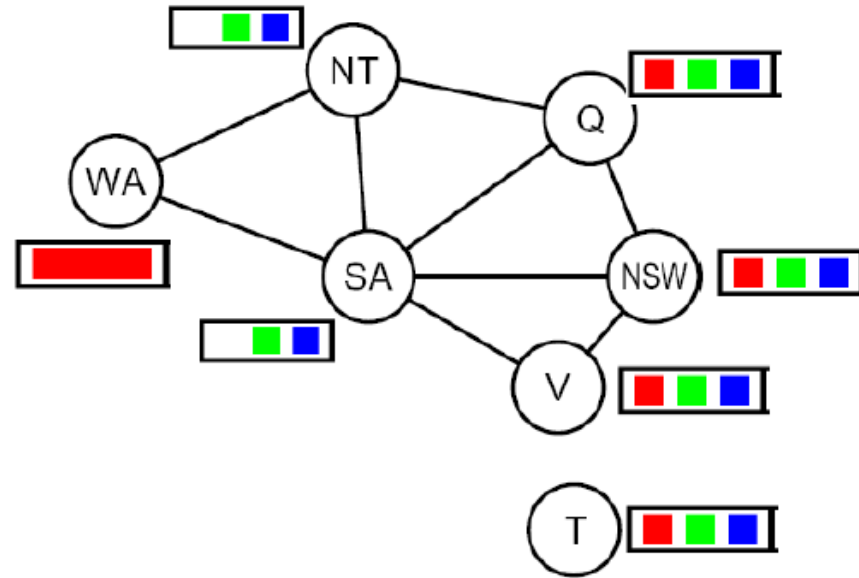
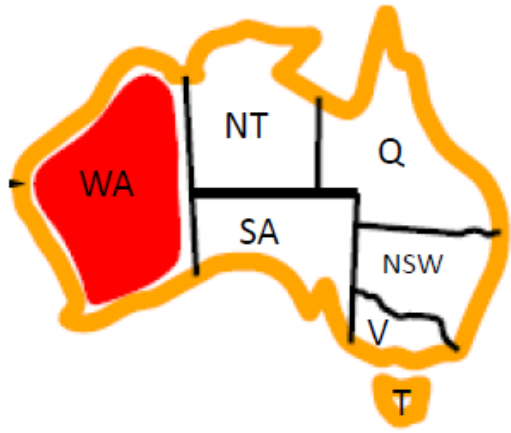
V->SA

WA->NT

SA->NT

Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

NT->SA

Q->SA

NSW->SA

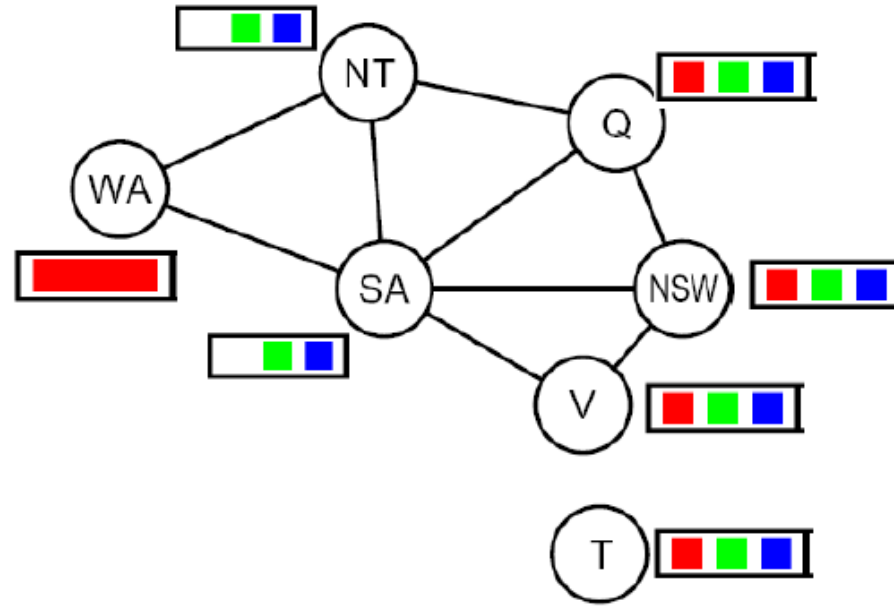
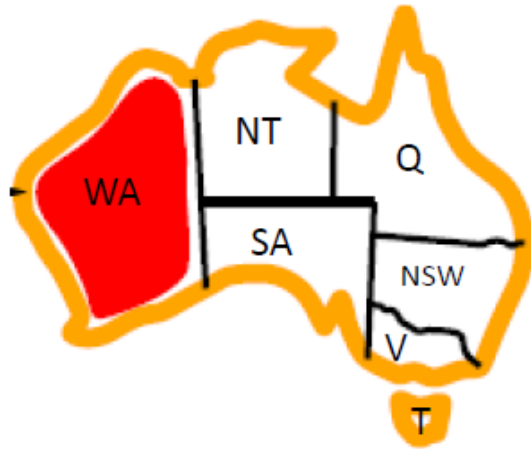
V->SA

WA->NT

SA->NT

Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

Q->SA

NSW->SA

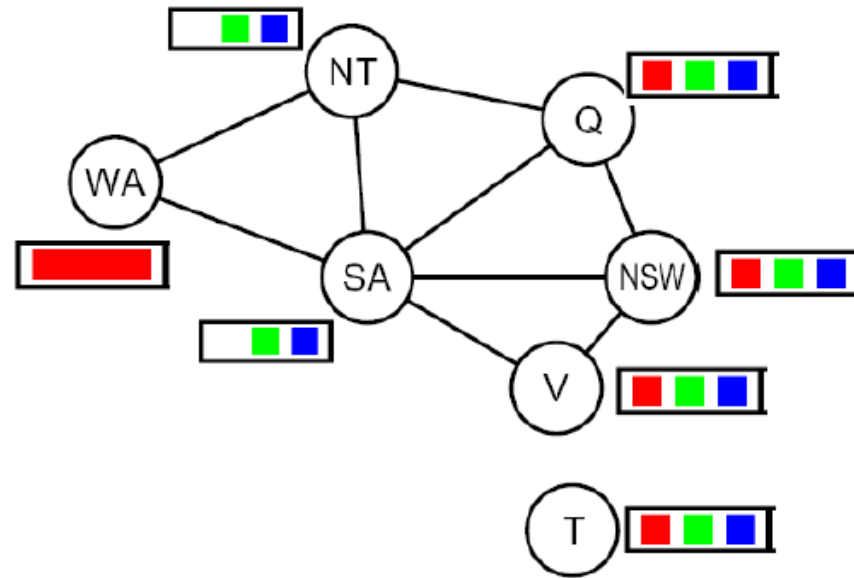
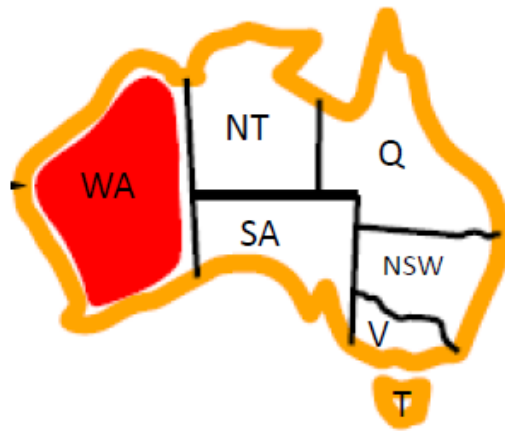
V->SA

WA->NT

SA->NT

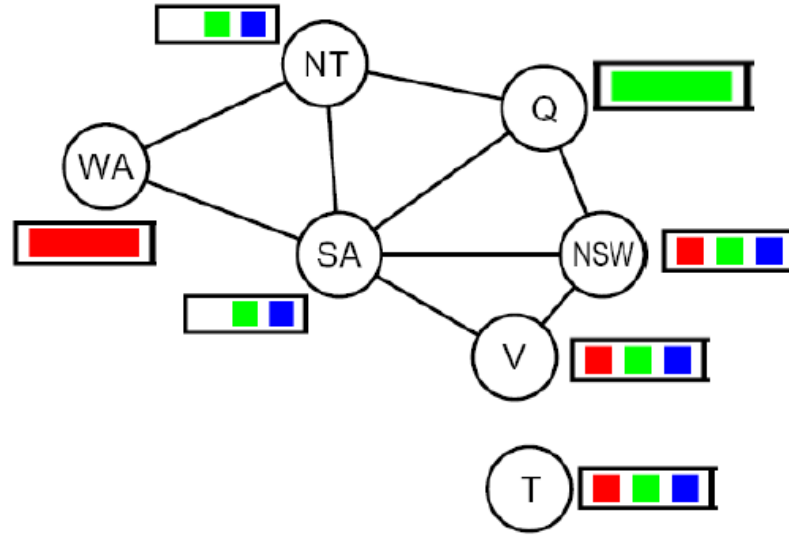
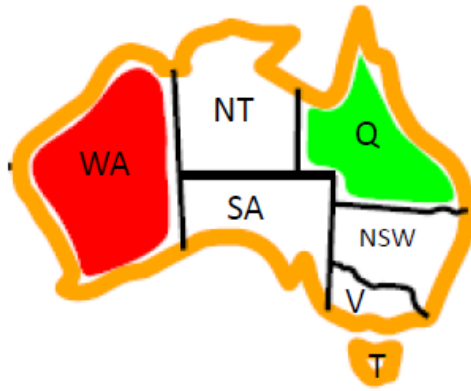
Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



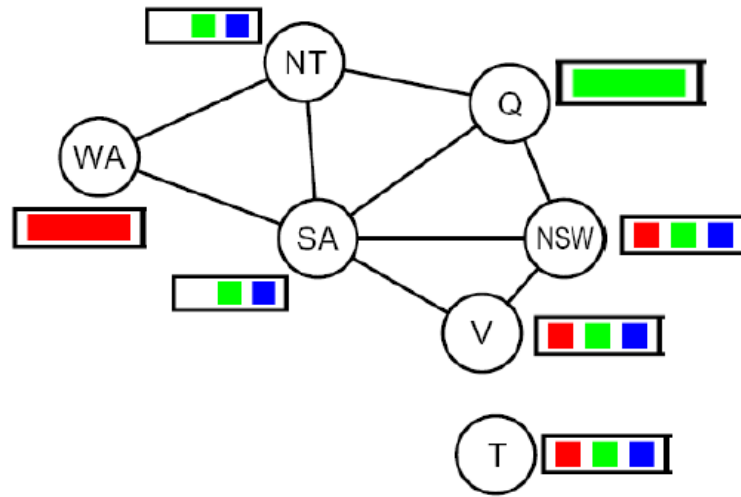
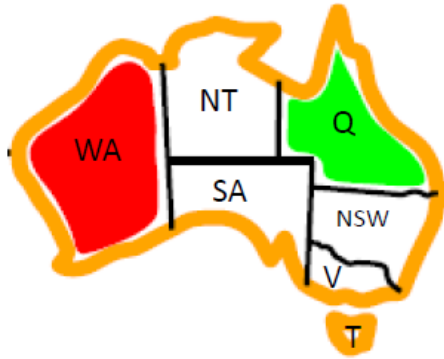
Queue:
NSW->SA
V->SA
WA->NT
SA->NT
Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

Piazza POLL: What gets added to the Queue?

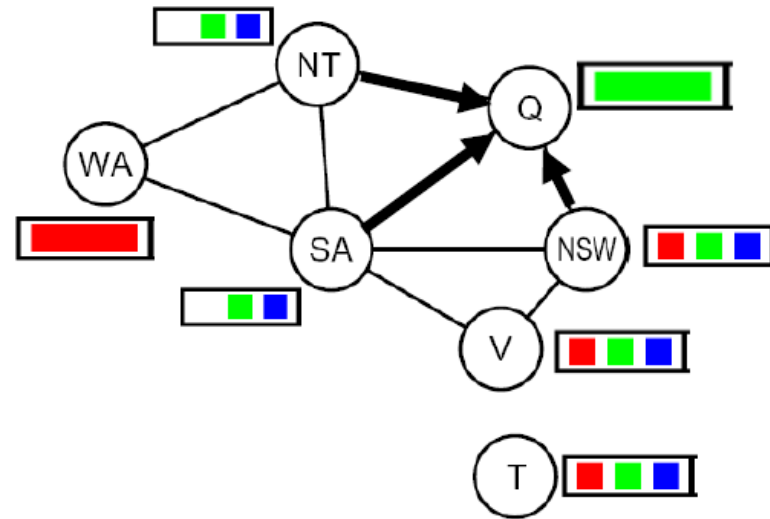
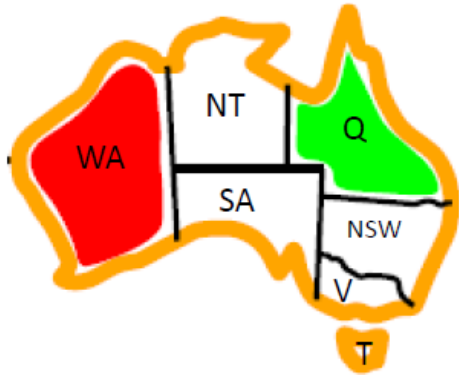


Queue:

A: NSW->Q, SA->Q, NT->Q

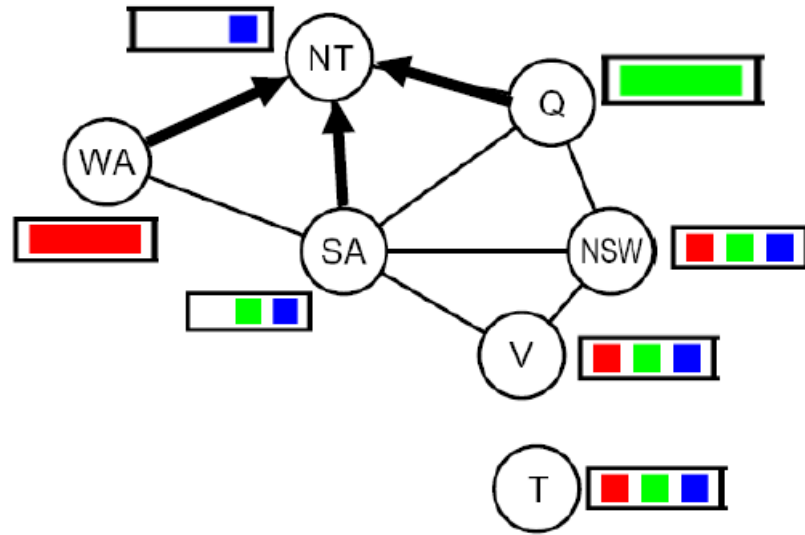
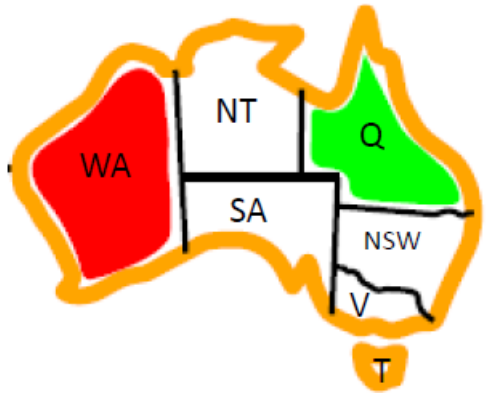
B: Q->NSW, Q->SA, Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



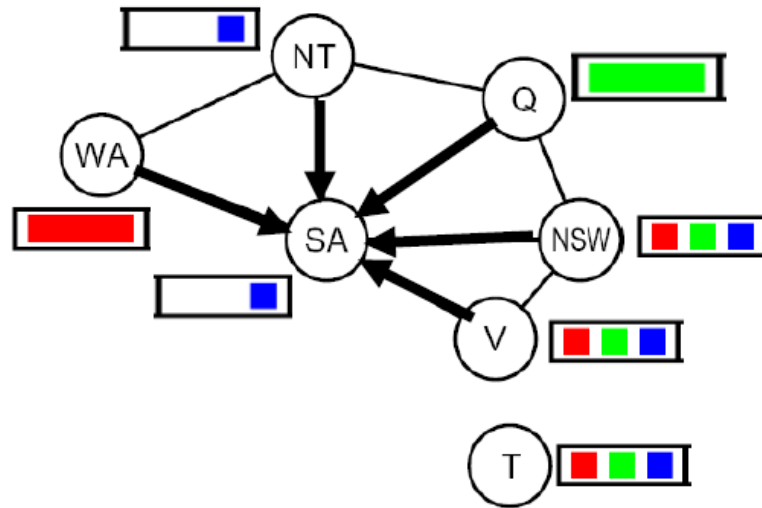
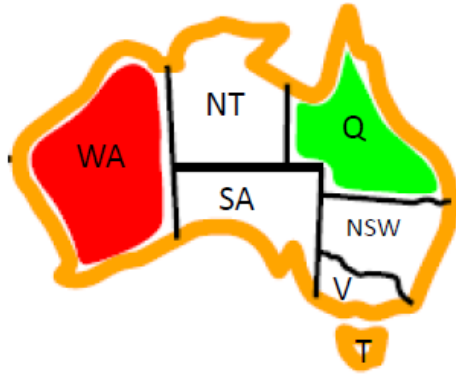
Queue:
NT→Q
SA→Q
NSW→Q

AC-3: Enforce Arc Consistency of Entire CSP



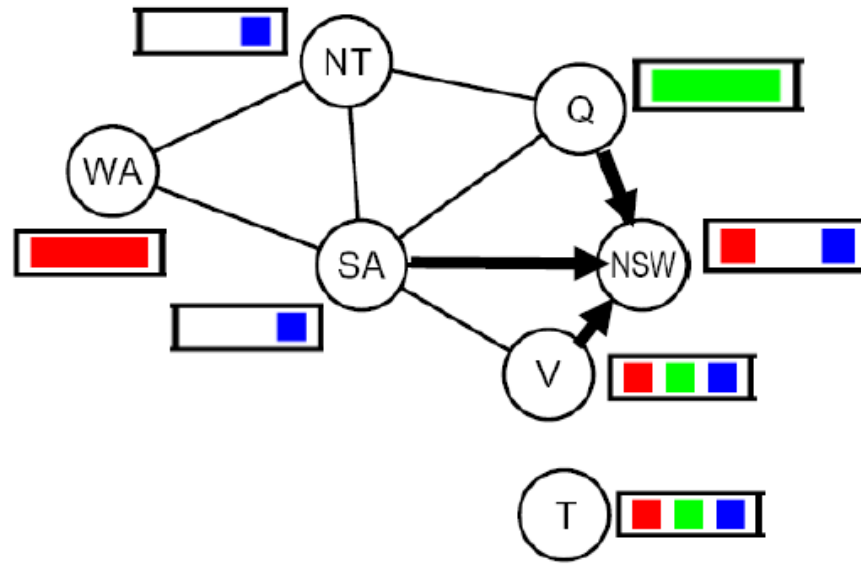
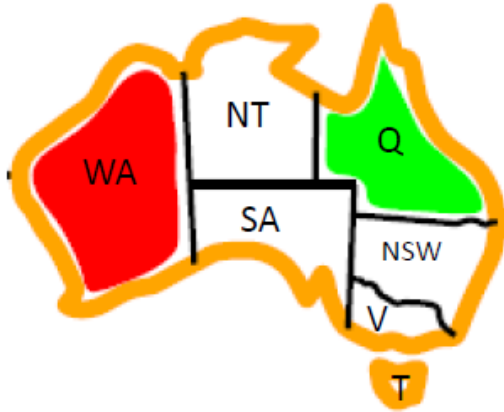
Queue:
SA->Q
NSW->Q
WA->NT
SA->NT
Q->NT

AC-3: Enforce Arc Consistency of Entire CSP



Queue:
NSW->Q
WA->NT
SA->NT
Q->NT
WA->SA
NT->SA
Q->SA
NSW->SA
V->SA

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

WA->NT

SA->NT

Q->NT

WA->SA

NT->SA

Q->SA

NSW->SA

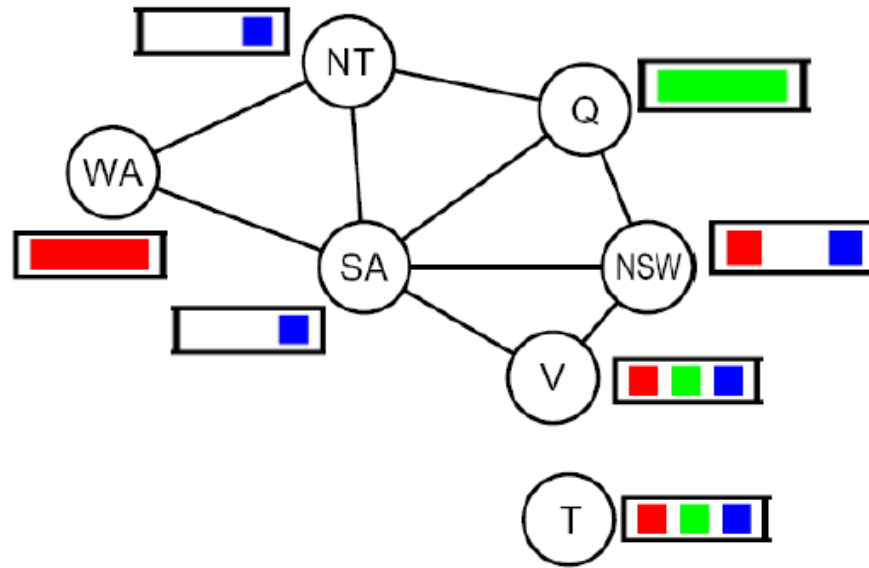
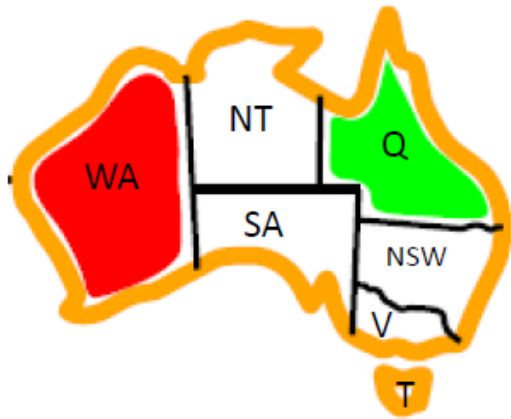
V->SA

V->NSW

Q->NSW

SA->NSW

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

WA->NT

SA->NT

Q->NT

WA->SA

NT->SA

Q->SA

NSW->SA

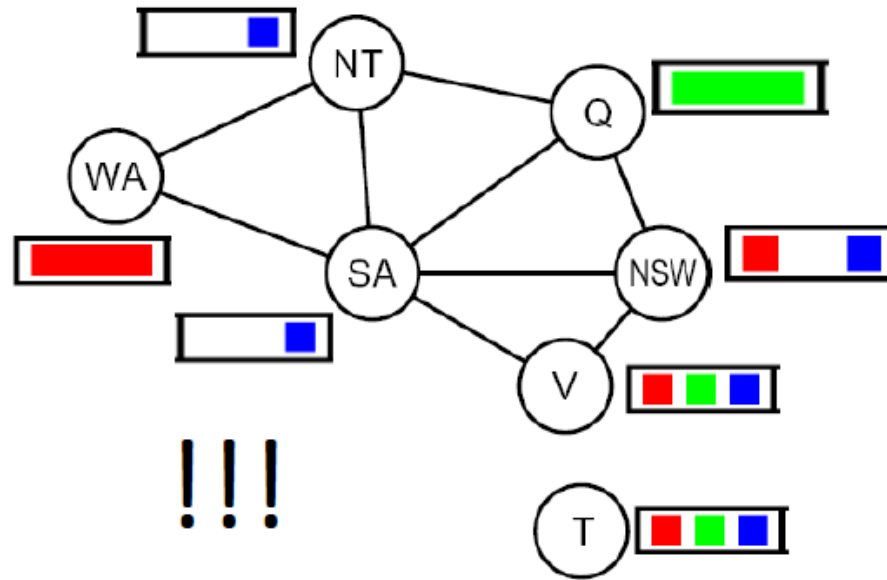
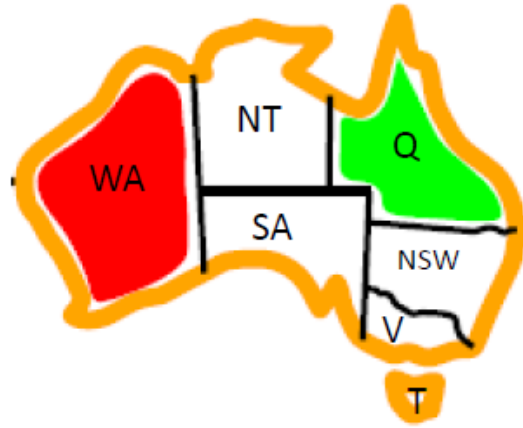
V->SA

V->NSW

Q->NSW

SA->NSW

AC-3: Enforce Arc Consistency of Entire CSP



Queue:

SA->NT

Q->NT

WA->SA

NT->SA

Q->SA

NSW->SA

V->SA

V->NSW

Q->NSW

SA->NSW